

Multiple Choice (10 marks)

1. Gwen wrote the number pattern below on a piece of paper. 1, 5, 9, 13 What are the next two terms in Gwen's pattern?
 - (a) 15, 17
 - (b) 15, 19
 - (c) 17, 19
 - (d) 17, 21
2. The Sojourn family went on a vacation. They started with 2000. *If they spent 150 each day*, which expression represents how much money they had after x days?
 - (a) $1850x$
 - (b) $2000 - 150x$
 - (c) $150x$
 - (d) $2000 + 150x$
3. Simplify $\frac{k-3}{2} + 3k + 1 + \frac{3k+1}{2}$.
 - (a) $3k+1$
 - (b) $5k$
 - (c) $3k$
 - (d) 1
4. Which step explains how to find the value of a in $6a = 72$?
 - (a) add 6 to both sides
 - (b) divide both sides by 6
 - (c) multiply both sides by 6
 - (d) subtract 6 from both sides
5. The rainfall for the last three months of school was 8.1 inches, 4.2 inches, and 0.33 inch. Estimate the total rainfall for the three months by rounding to the nearest whole number.
 - (a) 11.9 in.
 - (b) 11 in.
 - (c) 45.0 in.
 - (d) 12 in.

True / False (10 marks)

Answer True or False

6. True or False: The expression $9(9\text{ m} + 3\text{ t})$ simplifies to $18\text{ m} + 12\text{ t}$.
7. True or False: Sam would pay 5.52 *for 12 stamps based on the rate of* 8.28 for 18 stamps.
8. True or False: The ratio 12 over 27 can form a proportion with 8 over 18 because they are equivalent ratios.
9. True or False: The value of 'a' that makes the graphs of the equations perpendicular is -4.
10. True or False: The total cost of 934 puzzles at *6 each, including tax, is* 5,484.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A square is divided into five congruent rectangles, as shown. If the perimeter of each of these five rectangles is 36 inches, what is the perimeter of the square, in inches? [asy] draw((0,0)–(0,5)–(5,5)–(5,0)–cycle); draw((1,0)–(1,5)); draw((2,0)–(2,5)); draw((3,0)–(3,5)); draw((4,0)–(4,5)); [/asy]
12. Given that the mean of 7 numbers is 15, and upon adding an 8 th number the mean decreases to 12, calculate the 8 th number and explain your reasoning step-by-step.

End of Paper